

PAPER 1 / QUESTION 24

7 November 2024 / Zones A/B/C / HL Paper 1 / original question 4 / 8824-9590

- 24.** A balloon rises at a steady vertical velocity of 10 m s^{-1} . An object is released at rest relative to the balloon when the object is at a height 120 m above the ground. Air resistance is negligible.

What time, to the nearest second, does the object take to hit the ground?

- A. 4 s
- B. 5 s
- C. 6 s
- D. 12 s